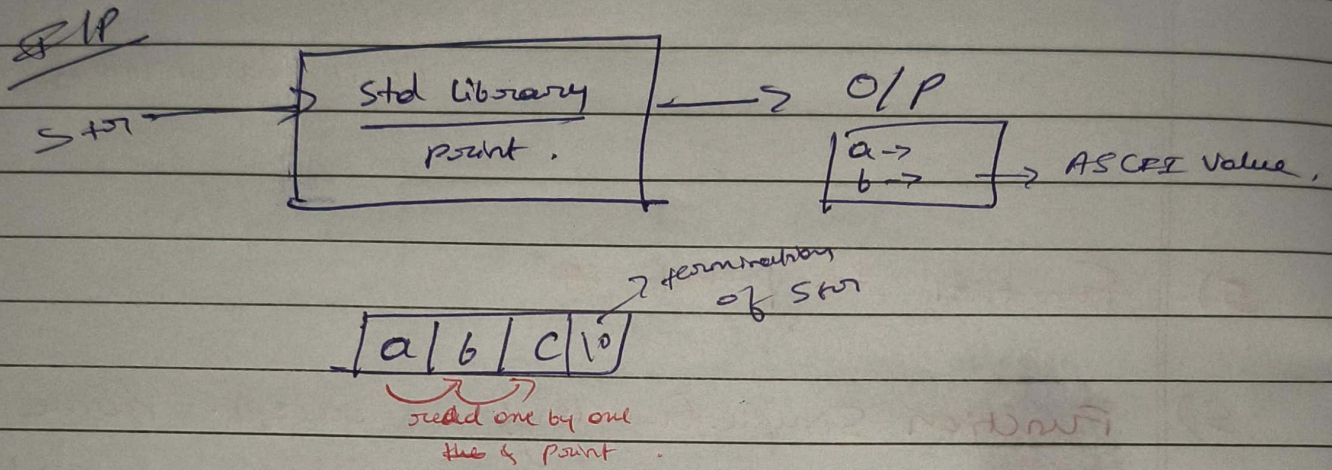
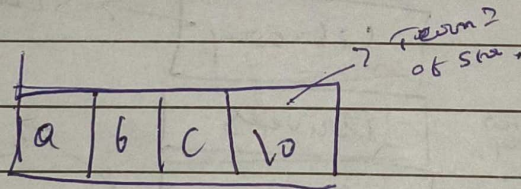


Coding Workshop - 2

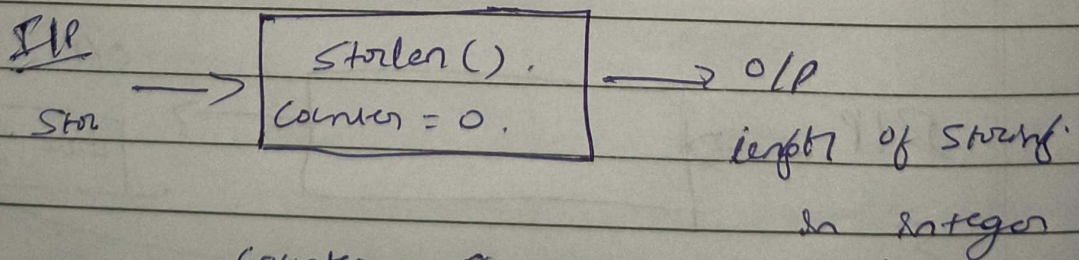
6) Function to print ASCII Values of String Input



7) Function to get String length to get string length



- ① Go over each char → loop
- ② Counter -
- ③ Str



Counter = Counter + 1

or,

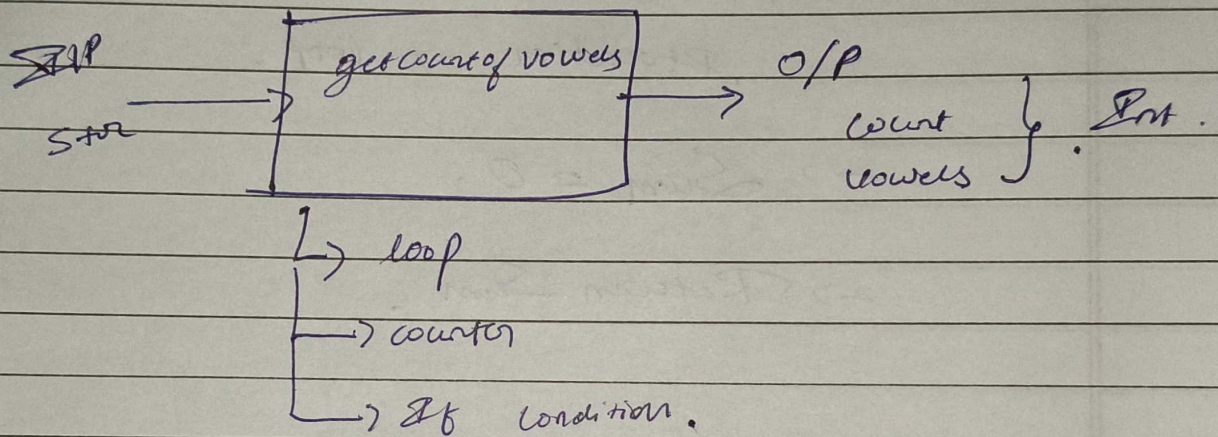
Counter ++ ;

8) function to Count vowels get Count Of O vowel for a given String.

a e i o u

A E I O U

use Counter



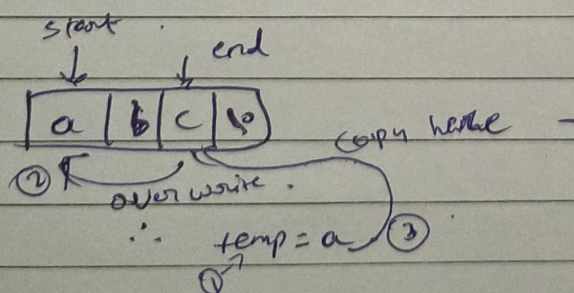
9) Function to Reverse String reverse String

Ex.

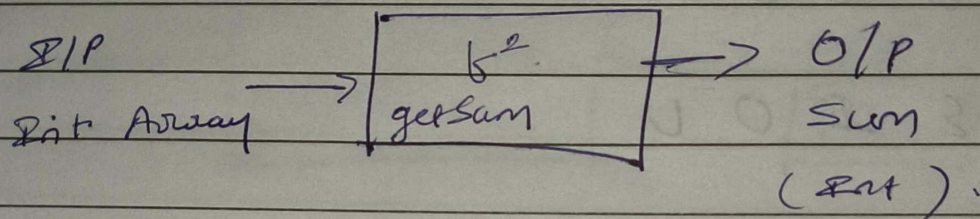
a	b	c	\0
---	---	---	----

c	b	a	\0
---	---	---	----

In place / change here only
or
In memory
no using extra str.



10) Function to get sum of all elements in the integer array getSum.



→ Read one value at a time
Iteration → loop.

→ Sum = 0.

→ Return Sum.

3_Coding_workshop_in_python > Coding_workshop_part1 > 6_ASCIIvalue.py > ...

```
1  def printASCIIofString(string):
2      for character in string:
3          print(f"{character} ->{ord(character)}") #ord is ASCII value finder
4
5  printASCIIofString("raj")
6
7
8  def printASCIIofString2(string2):
9      for character2 in string2:
10         ASCII_value = ord(character2)
11         print(f"{character2} ->{ASCII_value}") #ord is ASCII value finder
12
13  printASCIIofString("abc")
```

Coding_workshop_in_python > Coding_workshop_part1 > 7_strlength.py > str_length2

```
1 def str_length(string):  
2     counter = 0  
3     for character in string:  
4         counter += 1  
5  
6     return counter
```

```
7  
8 input = "None" #None -> null but running "None"  
9 print(f"string length of {input} is {str_length(input)}")
```

```
10  
11 input2 = ""  
12 print(f"string length of {input2} is {str_length(input2)}")
```

```
13  
14 input3 = "R"  
15 print(f"string length of {input3} is {str_length(input3)}")
```

```
16  
17 input4 = "raja"  
18 print(f"string length of {input4} is {str_length(input4)}")
```

```
19
20 # new for NOne IMPORTANT
21
22 def str_length2(string2):
23     counter = 0
24     if string2 != None:
25         for character2 in string2:
26             counter += 1
27
28     return counter
29
30 input = None
31 print(f"string length of {input} is {str_length2(input)}")
32
33 input2 = ""
34 print(f"string length of {input2} is {str_length2(input2)}")
35
36 input3 = "R"
37 print(f"string length of {input3} is {str_length2(input3)}")
38
39 input4 = "raja"
40 print(f"string length of {input4} is {str_length2(input4)}")
```

1st_2_numbers.py

6_ASCIIvalue.py

7_strlength.py

10_getsum.py

8_Vowels.py

3_Coding_workshop_in_python > Coding_workshop_part1 > 8_Vowels.py > count_vowels

```
1  def count_vowels(string)-> int:
2      counter = 0
3      if string != None:
4          for character in string:
5              if (character == 'a' or
6                  character == 'e' or
7                  character == 'i' or
8                  character == 'o' or
9                  character == 'u' or
10                 character == 'A' or
11                 character == 'E' or
12                 character == 'I' or
13                 character == 'O' or
14                 character == 'U' ):
15
16                 counter +=1
17
18     return counter
19 input = None
20 print(f"number of vowels in {input} is = {count_vowels(input)} ")
21
22 input = "raja"
23 print(f"number of vowels in {input} is = {count_vowels(input)} ")
24
25 input = "123"
26 print(f"number of vowels in {input} is = {count_vowels(input)} ")
```

```
20  
21  
22 def revers_string2(string2):
```

```
23  
24     length = len(string2)
```

```
25  
26     revers_string2 = ""
```

```
27     for end in range (length-1,-1,-1):
```

```
28         revers_string2 += string2[end]
```

```
29     return revers_string2
```

```
30  
31 name2 = "raja"
```

```
32 name2 = revers_string2(name2)
```

```
33 print(name2)
```

```
34  
35  
36 def reverse_string(name:str):
```

```
37  
38     if len(name) == 0:
```

```
39         return name
```

```
40     return reverse_string(name[1:]) + name[0]
```

```
41  
42 name = "Raja"
```

```
43 print(reverse_string(name))
```

```
44
```

1 #without range function

2 def getSum(numbers:list)-> int:

3
4 sum = 0

5
6 for number in numbers:

7 sum += number

8
9 return sum

0
1 numbers = [1,2,3]

2 sumOfNumbers = getSum(numbers)

3 print(f"sum of numbers = {sumOfNumbers}")

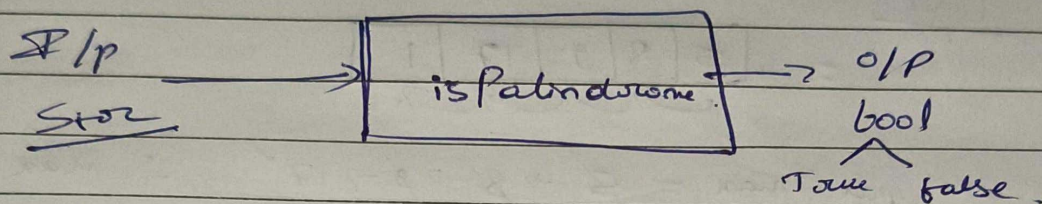
4

```
15
16 #with range function
17
18 def getSum1(numbers1:list)-> int:
19     sum1 = 0
20
21     length = len(numbers1)
22
23     for index in range(0, length, 1):
24         sum1 = sum1 + numbers1[index]
25
26     return sum1
27
28 numbers1 = [1,2,3,4]
29 sumOfNumbers1 = getSum1(numbers1)
30 print(f"sum of numbers = {sumOfNumbers1}")
```

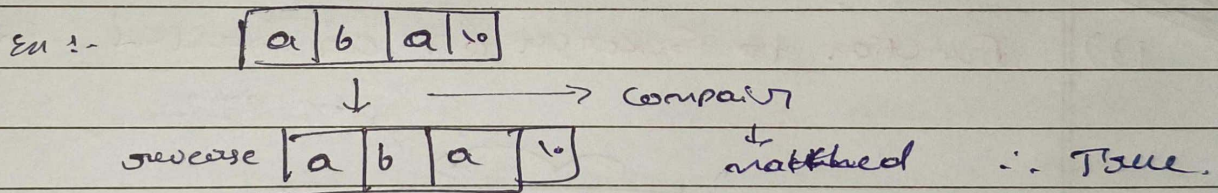
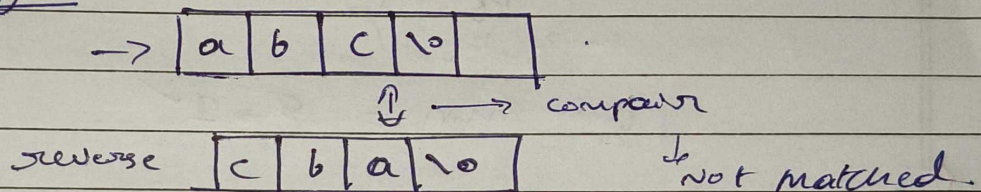
10) Function to find given string is palindrome, function `ispalindrome` returns true if string is palindrome.

Ex:- mom
→ mom

Ex:- malayalam
malayalam

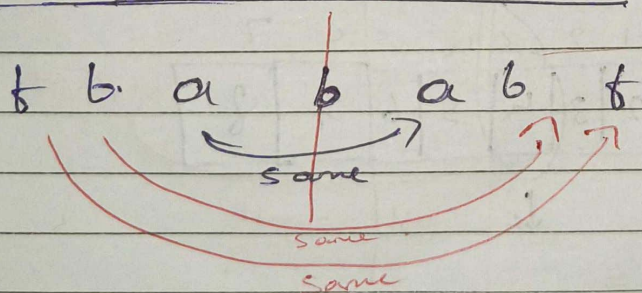


logic

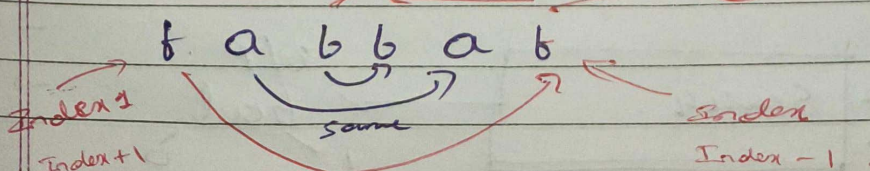


for No Extra memory :-

total no. 7 → odd No.

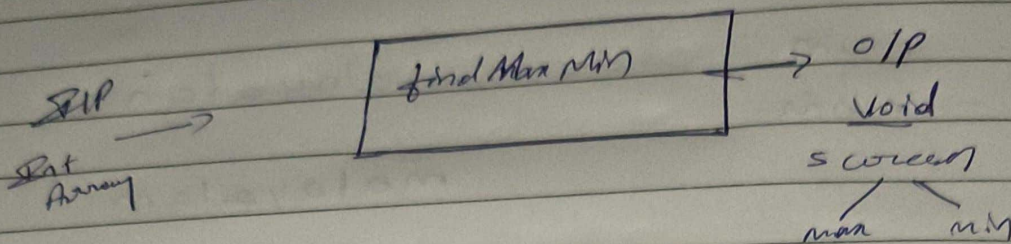


total no. 6 → even No.

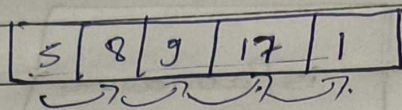


12)

Function to print max & min value from an integer array, printMaxMin.



Logic



for max = 5 → 8 → 9 → 17 ✓ ∴ Max = 17.

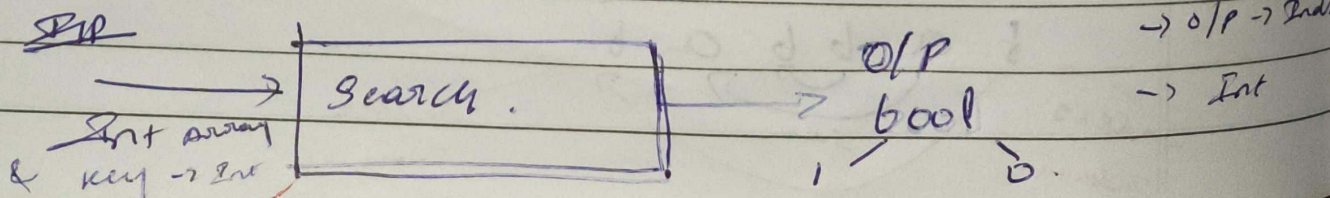
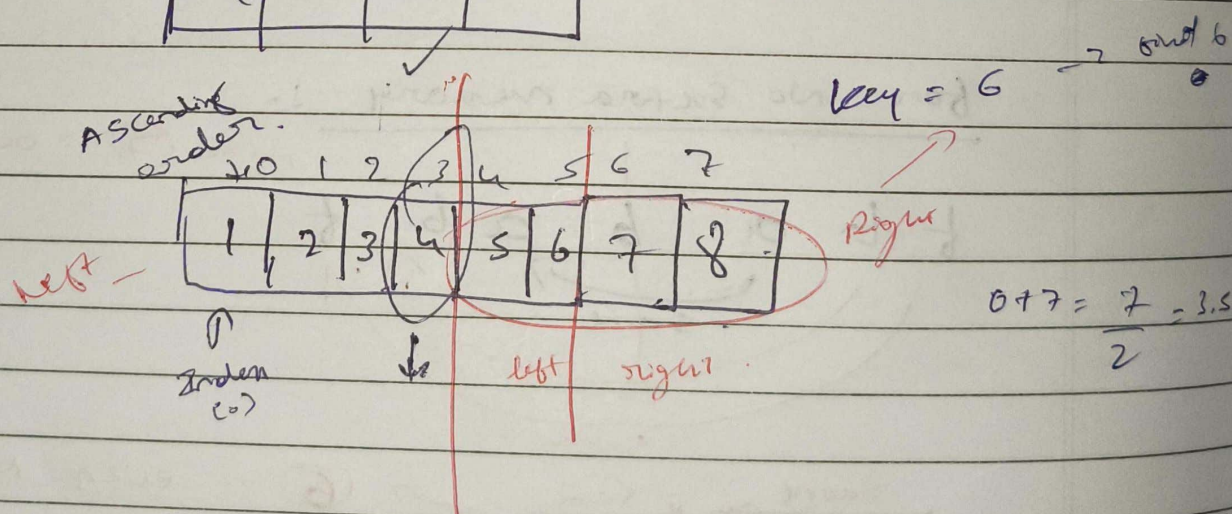
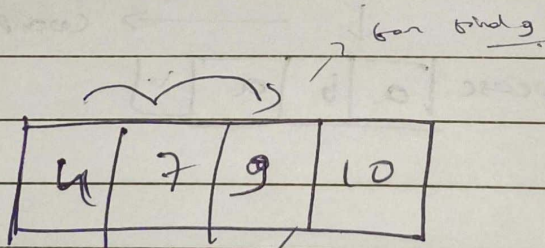
→ loop →

→ If ...
→ If ...

for min = 5 → 1 ✓

Imp

13) Function to search in an sorted integer array



```
3_Coding_workshop_in_python > Coding_workshop_part3 > 11_isPlaindrome.py > ...
1  #for not using extra memory
2
3  def isPalindrome(string:str)->bool:
4      leftIndex = 0
5      rightIndex = len(string) - 1
6      result = True
7
8      while (leftIndex < rightIndex):
9          if string[leftIndex] != string[rightIndex]:
10             result = False
11             break
12
13             leftIndex += 1
14             rightIndex -= 1
15
16     return result
17
18     input1 = "mom"
19     input2 = "malayalam"
20     input3 = "hello"
21     input4 = "abba"
22
23     result = isPalindrome(input4)
24
25     if result:
26         print("is palindrome")
27
28     else:
29         print("not palindrome")
```

3_Coding_workshop_in_python > Coding_workshop_part3 > 12_printMaxMin.py > ...

```
1  # def getMaxMin(numbers:list)-> tuple[int,int]:
2  def getMaxMin(numbers:list):
3      max = numbers[0]
4      min = numbers[0]
5
6      number = len(numbers)
7      for index in range(1,number,1):
8          if numbers[index] > max:
9              max = numbers[index]
10
11          if numbers[index] < min:
12              min = numbers[index]
13
14
15
16      return max,min
17
18 input1 = [0,2,113,4]
19
20 resultMax, resultMin = getMaxMin(input1)
21
22 print(f"max value is {resultMax} and min value {resultMin}")
```

11_isPlindrome.py

12_printMaxMin.py

13_searchNumber.py

3_isEven.py

3_Coding_workshop_in_python > Coding_workshop_part3 > 13_searchNumber.py > ...

```
3  def search(number, key):
4      leftIndex = 0
5      rightIndex = len(number) - 1
6      found = False
7
8      while(leftIndex <= rightIndex):
9          middleIndex = int(leftIndex + ( rightIndex - leftIndex /2 ))
10
11         if (number[middleIndex] == key):
12             found = True
13             break
14
15         if(number[middleIndex] > key):
16             rightIndex = middleIndex -1
17
18         else:
19             leftIndex = middleIndex + 1
20
21     return found
22
23
24     input1 = [1,2,3,4,5,6,7]
25     key = 7
26
27     result = search(input1, key)
28
29     if result:
30         print("number is found")
31
32     else:
33         print("number is not found")
```

PROBLEMS

OUTPUT

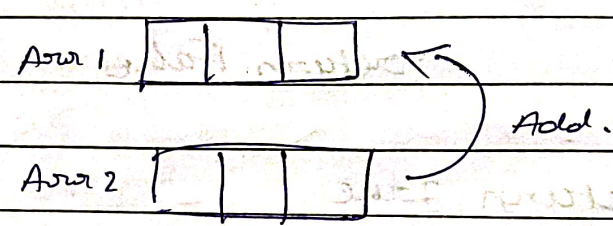
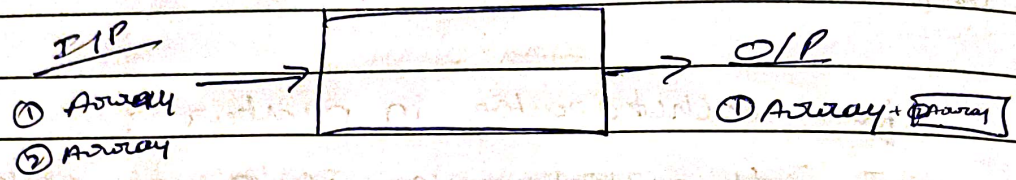
DEBUG CONSOLE

TERMINAL

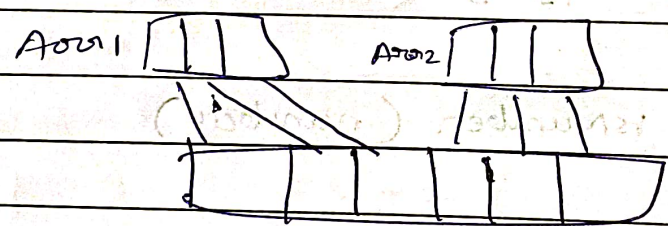
PORTS

Coding Workshop part - 4

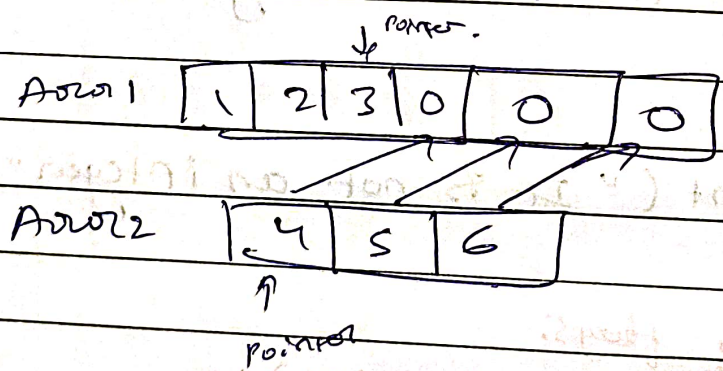
14) Function to merge two arrays & return Combined
o/p in first array



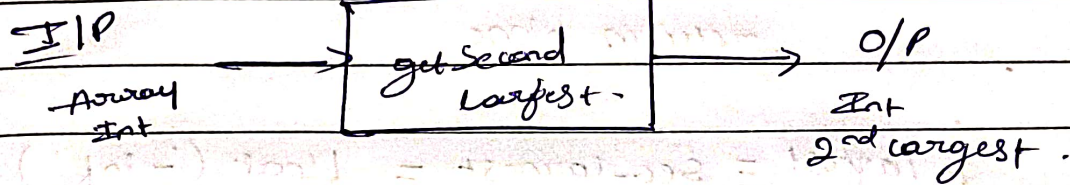
for No space in Array 1.



for Space in Array 1.



Function get Second largest element in an integer array get Second largest.



5, 9, 6, 3, 1, 2

1st largest

2nd largest

or.

Sort this.

1, 2, 3, 5, 6, 9

→ Max Large = 9 } assume 1st element is max.
→ Max Small = 6

~~Imp~~

```
def second-largest (arr):
```

```
    if len(arr) < 2:
```

```
        return None
```

```
    largest = sec-largest = float('-inf')
```

```
    for number in arr:
```

```
        if number > largest:
```

```
            sec-largest = largest
```

```
            largest = number
```

```
    elif number > sec-largest and number != largest:
```

```
        sec-largest = number
```

```
    if sec-largest == float('-inf'):
```

```
        return None
```

```
    return sec-largest.
```

```
arr = [8, 1, 3, 5, 9, 2]
```

```
result = second-largest(arr)
```

```
Print("second largest is", result).
```

3_Coding_workshop_in_python > Coding_workshop_part4 > 14_mergelist.py > ...

```
1  def mergeList(list1:list,list2:list)-> list:
2
3      for item in list2:
4          list1.append(item)
5
6
7
8  list1 = [1,2,3]
9  list2 = [14,5]
10
11  mergeList(list1,list2)
12
13  print(f"the list1 = {list1}")
14
15  #my own
16  def listmerge(list1,list2):
17      return list1 + list2
18
19  list1 = [1,2,3]
20  list2 = [4,5]
21
22  result = listmerge(list1,list2)
23  print("list merge is =",result)
```

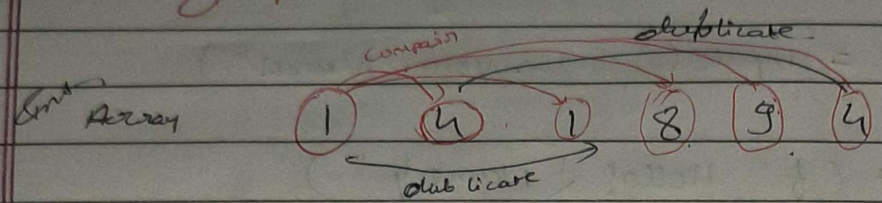
Practice_Coding > 2ndlargest.py > max_val

```
1 def second_largest(arr):
2
3     if len(arr) < 2:
4         return None
5
6
7     largest = seclargest = float('-inf')
8
9     for number in arr:
10         if number > largest:
11             seclargest = largest
12             largest = number
13
14         elif number > seclargest and number != largest:
15             seclargest = number
16
17     if seclargest == float('-inf'):
18         return None
19     return seclargest
20
21 arr = [2,8,9,1,0]
22
23 result = second_largest(arr)
24
25 print("secend largest is ",result)
26
27
28 # small example
```

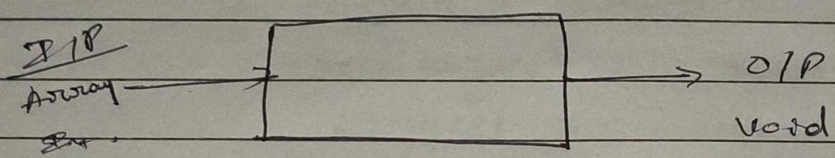
B Coding Workshop - part 5

Ques

16) Function to print Unique elements in an integer array printUniqueElements.



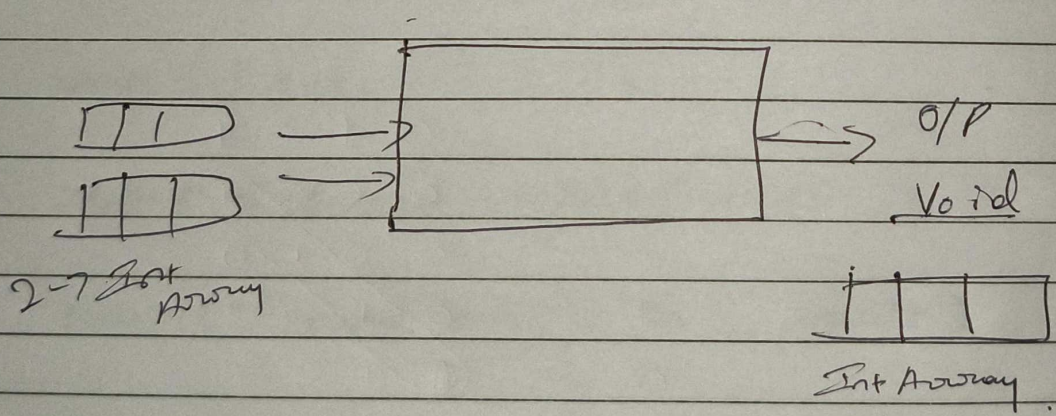
Unique elements :- 1, 4, 8, 9, 1



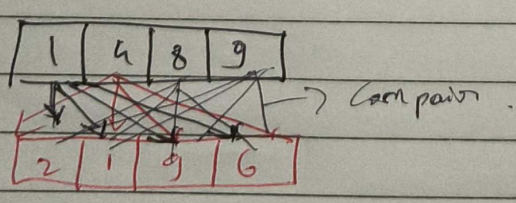
print -> distinct elements.

Ques
Common Element

17) Function to print Intersection or Common elements of two integer array getCommonElements.



not necessary in C++
but also instead of this return it -> int.



① ⑨ -> Return / Print 74.

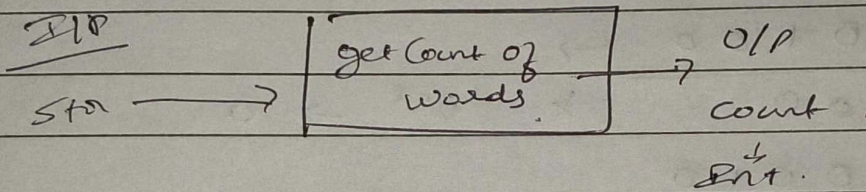
18) Function to get count of words in the string
getCountOfWords

Ex: How are you
1 2 3

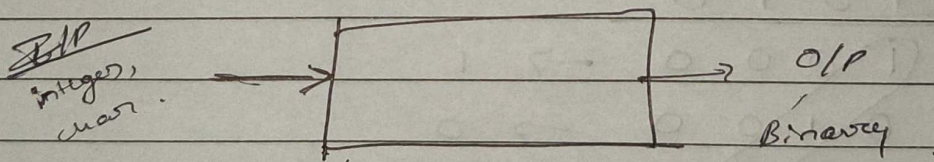
Count of No. 3

counter $\rightarrow$ space . $++$ $++$ $++$

Count $1 + 1 + 1 = 3$



19) Function to print binary values of various input like Integer, char, also performs shift operations on Integer.



Left $\leftarrow$ Right

0	0	0	0
1	0	1	0

$\times (2^3 \quad 2^2 \quad 2^1 \quad 2^0) \times$

8 4 2 1

8 + 2 + 0 = 10

bind 1 on 0 here

1	0	1	0
---	---	---	---

places

Input $\rightarrow$ 1 0 0 0

1	0	0	0
---	---	---	---

$\rightarrow$ bitwise & (multiples)

works as mask (when it on mem 1 then perform operation and then we can easily find the another behind that)

A	B	A & B
1	0	0
1	1	1
0	0	0
0	1	0

Ex:-

$$\begin{array}{r} 1010 \\ 1000 \end{array} \&$$

$$\begin{array}{r} 1000 \\ 0100 \end{array} \&$$

$$\begin{array}{r} 0100 \\ 0000 \end{array} \&$$

$$\begin{array}{r} 0010 \\ 0000 \end{array} \&$$

$$\begin{array}{r} 0001 \\ 0000 \end{array} \&$$

→ It comes +ve. & find the left most 1's
 → 2nd → 0
 → 3rd → 1
 → 4th → 0

$$\begin{array}{r} 1010 \\ 1000 \end{array} \rightarrow 1$$

$$\begin{array}{r} 0100 \end{array} \rightarrow 0$$

$$\begin{array}{r} 0010 \end{array} \rightarrow 1$$

$$\begin{array}{r} 0000 \end{array} \rightarrow 0$$

32
 1 | 0 | 0 | 0 | 1 |

Unsigned int a = 1
 Internally → 1010 → 11 → 10

32 times jump
 left → right
 1 → 1st pos

unsigned int $n = 1$

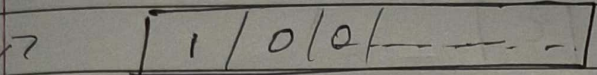
$$n = n \ll 31$$

Size of

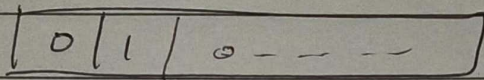
gives

$$4 \times 8 = 32 - 1 = 31$$

↓
bytes



Shift 1 bit



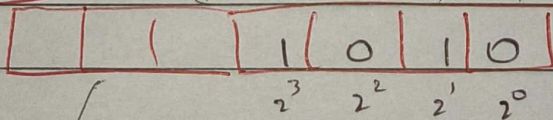
for -ve :- 2^s Compliments.

$$\begin{array}{r} 1000 \rightarrow 10 \\ \underline{} \\ 0101 \end{array} \rightarrow \text{gives: means } 0 \rightarrow 1$$

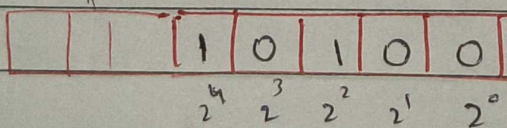
then + $ + 1$

2^s Compliment $\rightarrow 0000 \rightarrow -10$

for left Shift Ex:-



$$8 + 0 + 2 + 0 = 10$$



$$16 + 0 + 4 + 0 + 0 = 20$$

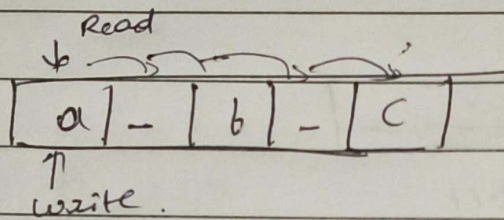
for left Shift by 1

* 2 bit multiple by 2

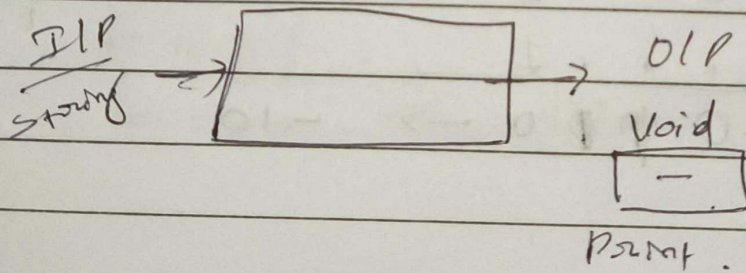
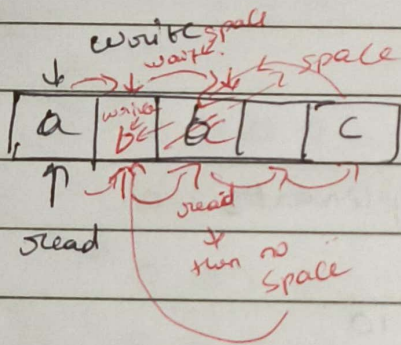
for output Shift
it will be divided by
2.

20) Function to remove spaces from the string
removeSpaces.

a b c $\rightarrow$ abc
~~ab~~
 ✓
 Remove



Sample



coding_workshop_in_python > Coding_workshop_part5 > 16_duplicate_number.py

```
def print_Unique_Number(numbers):  
    for read_index in range(0, len(numbers), 1):  
        isDuplicate = False  
  
        for compair_index in range(0, len(numbers), 1):  
            if read_index == compair_index:  
                continue  
  
            elif numbers[read_index] == numbers[compair_index]:  
                isDuplicate = True  
                break  
  
        if isDuplicate == False:  
            print(numbers[read_index]) #this the the unique element hence print it  
  
numbers = [1,4,5,4]  
print_Unique_Number(numbers)
```

coding_workshop_in_python > Coding_workshop_part5 > 17_common_friend.py > getCom

```
1 def getCommonElement(input1 , input2):
2
3     for input1_index in range(0, len(input1), 1):
4         isFound = False
5
6         for input2_index in range(0, len(input2), 1):
7
8             if input1[input1_index] == input2[input2_index]:
9                 isFound = True
10                break
11
12            if isFound:
13                print(input1[input1_index])
14
15 input1 = [1,2,3,4,1]
16 input2 = [3,2,5,8,9,1]
17
18 getCommonElement(input1, input2)
19
```

5_coding_workshop_in_python > Coding_workshop_part5 > 18_WordCount.py > ...

```
3 def getWordCount(String):
4     counter = 0
5
6     for char in String:
7         if char == ' ' or char == '\t' or char == '\n':
8             counter +=1
9
10    # spacial case with just one word
11
12    if (len(String) > 0 and counter == 0):
13
14        return 1
15
16    return counter
17
18 String = ""
19
20 result = getWordCount(String)
21
```

tsNumber.py 19_printInttoBinary.py x 20_removeSpace.py 17_con

Coding_workshop_in_python > Coding_workshop_part5 > 19_printInttoBinary.py > pri

```
13 def printBinaryForm(number:int):
14
15     no_of_bits = number.bit_length()
16
17     print(f"Number of bits {no_of_bits}")
18
19     mask = 1
20     mask = mask << no_of_bits # left bitwise oprator
21
22     for _ in range(no_of_bits):
23         if number & mask:
24             print("1", end = "") # end = "" this is for printing num
25         else:
26             print("0", end = "")
27
28         mask = mask >> 1 # right shift bitwise oprator
29 printBinaryForm(1024)
```

PROBLEMS OUTPUT

Coding_workshop_in_python > Coding_workshop_part5 > 20_removeSpace.py > removeSpaceString

```
1 def removeSpaceString(string):  
2     output = ''  
3  
4     for character in string:  
5         if character != ' ' and character != '\t' and character != '\n':  
6             output = output + character  
7  
8     print(output)  
9  
10 input = "a"  
11 removeSpaceString(input)
```